

Article

Enhancing Wireless Sensor Networks with Bluetooth Low-Energy Mesh and Ant Colony Optimization Algorithm

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Abstract

Wireless Sensor Networks (WSNs) face persistent challenges of uneven energy depletion, limited scalability, and reduced network lifetime, all of which hinder their effectiveness in Internet of Things (IoT) applications. This paper introduces a hybrid framework that integrates Bluetooth Low-Energy (BLE) mesh networking with Ant Colony Optimization (ACO) to deliver energy-aware, adaptive routing over a standards-compliant mesh fabric. BLE mesh contributes a resilient many-to-many topology with Friend/Low-Power Node roles that minimize idle listening, while ACO dynamically selects next hops based on residual energy, distance, and link quality to balance load and prevent hot spots. Using large-scale simulations with 1000 nodes over a 1000×1000 m field, the proposed BLE-ACO system reduced overall energy consumption by approximately 35%, extended network lifetime by 40%, and improved throughput by 25% compared with conventional BLE forwarding, while also surpassing a LEACH-like clustering baseline. Confidence interval analysis confirmed the statistical robustness of these results. The findings demonstrate that BLE-ACO is a scalable, sustainable, and standards-aligned solution for energy-constrained IoT deployments, particularly in smart cities, industrial automation, and environmental monitoring, where long-term performance and adaptability are critical.

Keywords: BLE; WSN; ACO; energy efficiency; IoT

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1. Introduction

Wireless Sensor Networks (WSNs) underpin modern IoT applications, including environmental monitoring, industrial automation, and smart infrastructure, yet tight energy budgets and uneven battery depletion still constrain longevity; classical clustering, such as Low-Energy Adaptive Clustering Hierarchy (LEACH), reduces control overhead but can create hot traffic spots near cluster heads, accelerating node failures and shortening network lifetime [1]. In parallel, Bluetooth Low-Energy (BLE) and its mesh capabilities (introduced with Bluetooth 5.x) offer a promising substrate for large-scale, battery-powered sensing: BLE is designed for ultra-low-power operation [2], while BLE mesh provides many-to-many communication with decentralized, self-healing behavior and Friend/Low-Power Node (LPN) roles that curb idle listening features well suited to dynamic, reliability-critical deployments in smart cities and industrial settings [3]. Figure 1 emphasizes the two pillars it targets: dynamic path determination and real-time adaptation, motivated by recent evidence that BLE-mesh reliability and latency are sensitive to parameters such as

transmit power, packet/advertising interval, relay density, and overall network structure, underscoring the need for adaptive routing rather than static forwarding.

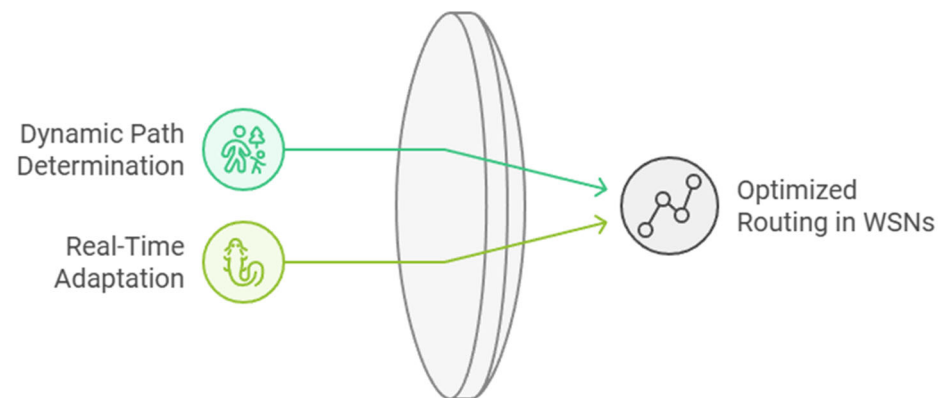


Figure 1. Enhancing WSN efficiency with ACO-driven, real-time adaptation.

Figure 2 introduces our hybrid approach that integrates BLE mesh with Ant Colony Optimization (ACO) to address these challenges. BLE mesh contributes a scalable, low-power communication fabric with Friend/Low-Power Node roles that reduce idle listening, while ACO supplies bio-inspired, energy-aware next-hop selection using residual-energy and distance heuristics [4]. The design targets the following five objectives: (i) minimize total energy consumption across the network; (ii) extend node-level and network-wide lifetime; (iii) improve scalability for large deployments; (iv) raise reliability and fault-tolerance via adaptive/multipath routing; and (v) optimize data delivery while maintaining low latency [5–7]. Comparative literature also indicates that, alongside ZigBee-like (IEEE 802.15.4) baseline and LoRa, BLE mesh is a competitive, low-cost choice for dense indoor IoT where short-range multi-hop and strong interoperability are desired [8].

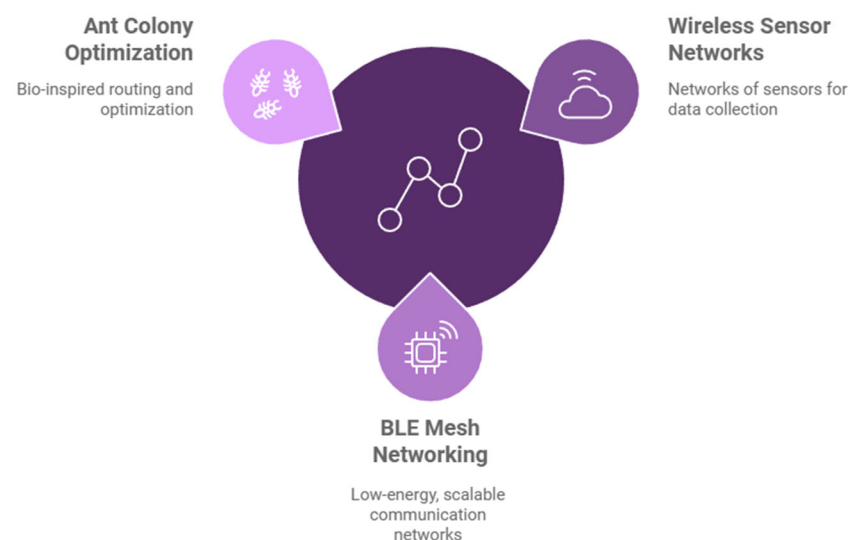


Figure 2. Integration of WSNs, BLE mesh, and ACO.

Most BLE-mesh systems rely on managed flooding or quasi-static paths, according to the Bluetooth Mesh specification and common tutorials on Time-to-Live (TTL)/relay-based managed flooding, and even where opportunistic forwarding is explored, forwarding decisions are seldom explicitly residual-energy-aware across the mesh [8,9]. It contributes a BLE-ACO hybrid that replaces pure flooding with pheromone-guided, energy-aware next-hop selection on standards-compliant BLE mesh to balance load and improve per-packet

energy efficiency. Contributions include the following: (i) a unified BLE-ACO routing framework with explicit heuristic factors and consistent notation; (ii) baselines beyond “traditional BLE,” including a LEACH-like clustering scheme to surface hot-spot effects; (iii) parameter-sensitivity analysis for α , β , and ρ with recommended stable ranges; and (iv) a reproducible simulation design (source/destination selection, traffic generation) with ≥ 10 runs and 95% confidence intervals. These choices are grounded in evidence that BLE-mesh performance is sensitive to TX power, packet/advertising interval, relay density, and network structure, motivating adaptive (energy-aware) routing, and in long-standing critiques of LEACH’s uneven energy depletion and hot-spot formation [8].

2. Integrating Bluetooth Low-Energy Mesh with Ant Colony Optimization

The proposed system is BLE mesh networking with ACO to deliver a scalable, energy-efficient, and self-organizing WSN that addresses core challenges of energy consumption, coverage, and fault tolerance. On the networking side, BLE mesh employs managed flooding with per-packet TTL, message caching, and relay roles to ensure delivery across multiple hops even when individual relays fail; Friend nodes buffer traffic for LPNs so sleeping devices can wake, poll, and receive queued messages, an approach that substantially reduces idle listening and extends battery lifetime [8]. Its security architecture provides authenticated encryption in the network/transport layers with distinct keys (NetKey/AppKey), key refresh, and replay protection, making BLE mesh attractive for large sensor deployments requiring resilience and commodity interoperability [8]. On the routing side, ACO complements BLE mesh by performing online, energy-aware next-hop selection: lightweight forward/return ants update pheromones, so paths adapt to residual energy, hop progress, and link quality, dynamically balancing load to avoid hot spots and premature node death [9]. Compared with alternatives such as Genetic Algorithms (GA) and Particle Swarm Optimization (PSO), ACO’s distributed, memory-light updates map well to duty-cycled nodes and small payloads while retaining strong convergence in WSN/IoT routing. Together, these traits yield a standards-compliant BLE-ACO stack that replaces pure flooding with pheromone-guided forwarding to improve per-packet energy efficiency without sacrificing robustness. BLE mesh networking enables many-to-many device communication, greatly expanding coverage and scalability while eliminating single points of failure through a self-healing, peer-to-peer topology. A mesh node can send, receive, and relay messages; within each node, elements act as functional units, and models specify standardized behaviors or services to ensure interoperability [8,10]. Specialized roles further improve efficiency: relay nodes re-transmit packets to extend range, and LPNs conserve energy by sleeping aggressively while a paired Friend node buffers messages for later retrieval, reducing idle listening and prolonging battery life [11]. At the protocol core, the network layer defines the Network PDU, applies encryption and authentication, and manages forwarding; BLE mesh employs managed flooding in which forwarding is constrained by TTL and aided by caches to prevent redundant retransmissions [11]. This robust architecture supports dynamic path availability and maintains reliable delivery even amid node failures, making BLE mesh well-suited to large-scale IoT, smart-home, and industrial deployments that demand reliable, scalable, and energy-efficient communication [12]. In our system, it retains standards-compliant managed flooding for robustness but replaces pure flooding at relays with ACO-guided next-hop selection, using local heuristics to steer traffic, balance load, and further reduce energy consumption while preserving the BLE mesh’s security and interoperability advantages [12,13].

ACO-Based Routing Protocol

Let $N(i)$ denote the neighbors of node i . It adopts an ACO-based routing protocol that begins with an initialization phase in which each feasible link (i, j) is assigned a common pheromone level $\tau_{ij} = \tau_0$, after which nodes compute path-selection probabilities using both pheromone intensity and a composite heuristic that encodes progress and energy awareness. Specifically, it defines γ_{ij} , the heuristic information for link (i, j) . It is a composite metric that combines distance, residual energy, and link quality [12,13]:

$$\gamma_{ij} = \frac{1}{d_{ij}} \cdot \frac{E_j}{E_{max}} \cdot LQI_j \quad (1)$$

where d_{ij} is the distance from i to j , $\frac{E_j}{E_{max}}$ is the normalized residual energy at node j , and $LQI_j \in (0, 1)$ is an optional link-quality indicator (set $LQI_j = 1$, when unavailable). P_{ij} The probability of selecting node j as the next hop from node i . Defined by the balance of pheromone trail and heuristic information [12]:

$$P_{ij} = \frac{\tau_{ij}^\alpha \cdot \gamma_{ij}^\beta}{\sum_k \tau_{ik}^\alpha \cdot \gamma_{ik}^\beta} \quad (2)$$

where τ_{ij} represents the pheromone level and α and β are the parameters that control the relative importance of pheromone and heuristic factors in path selection. Following each routing cycle, pheromone levels are updated to reflect the path quality, promoting energy-efficient paths and balancing the load across the network. The pheromone update rule $\Delta\tau_{ij}$ is deposited on the link (i, j) , proportional to the quality of the chosen path (e.g., inverse of total energy consumption or hop count) [12,14,15]:

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} + \Delta\tau_{ij}, \quad \Delta\tau_{ij} \propto \frac{1}{PathCost} \quad (3)$$

where $\rho \in (0, 1)$. Here, $PathCost$ is defined as the energy per successfully delivered bit (including retransmissions), so that lower-energy routes earn larger deposits $\Delta\tau_{ij}$. This design naturally favors higher-energy relays (through $\frac{E_j}{E_{max}}$), prefers better links (through LQI_j), and reduces hop count/distance (through $\frac{1}{d_{ij}}$), yielding online adaptation suitable for duty-cycled, resource-constrained WSN nodes while maintaining simple, distributed computations.

3. Mathematical Models and Optimization

The ACO algorithm is designed to minimize network-wide energy consumption while preserving reliable end-to-end delivery in the BLE-mesh fabric. It models per-packet transmission energy on the link (i, j) by a cost C_{ij} (J/bit) and expresses node i per-round expenditure as [6,13,15]:

$$E_i = \sum_{j \in N(i)} n_{ij} C_{ij} L \quad (4)$$

where n_{ij} is the number of packets forwarded on (i, j) and L is the payload in bits [15]. The objective of the optimization is to minimize the total energy expenditure across the network while ensuring reliable data transmission. The energy efficiency is thus formulated as [12,13]:

$$EnergyEfficiency = \min_i \sum_i E_i \quad (5)$$

In addition to energy optimization, the system also aims to extend the network lifetime. The lifetime of node i is determined by:

$$T_i = \frac{E_i}{P_i} \quad (6)$$

where E_i is the remaining energy of the node i , and P_i is the energy consumption rate [15]. Therefore, the energy saving refers to the reduction in total energy consumption achieved by the BLE mesh network integrated with the ACO algorithm compared to conventional BLE-based WSNs. It quantifies the efficiency gained through optimized routing paths that minimize redundant energy usage. Energy savings can be calculated as the percentage difference between the energy consumed by the traditional BLE system and the BLE-ACO system, using the following formula:

$$EnergySaving(\%) = \left(\frac{E_{traditional} - E_{BLE-ACO}}{E_{traditional}} \right) \times 100 \quad (7)$$

where $E_{traditional}$ is the total energy consumption of the traditional system, and $E_{BLE-ACO}$ is the energy consumption of the BLE-ACO configuration [6,13]. In practice, BLE mesh contributes resilience and interoperability through managed flooding with TTL, caching, relay roles, and Friend/LPN buffering, while ACO supplies lightweight, online, energy-aware next-hop adaptation that reduces redundant transmissions and balances forwarding load; together these mechanisms drive down $\sum_i E_i$, extend lifetimes T_i , and maintain delivery guarantees in dense, large-area IoT deployments.

4. Simulation Setup and Results

The simulation begins with a structured initialization phase in which network and algorithm parameters are carefully defined to evaluate the BLE-mesh WSN under ACO-based routing. A total of 1000 sensor nodes are uniformly distributed across a 1000×1000 m field, each initialized with 10 J of energy. Communication ranges vary between 10 and 50 m, the forwarding horizon is limited by $TTL = 10$, and experiments are conducted for 500 rounds. Nodes are assigned initial pheromone values (τ_0) and energy capacity, while duty cycles adapt dynamically to residual energy levels to minimize idle listening. BLE-specific roles are incorporated, with relays extending multi-hop connectivity and LPNs paired with Friend nodes for buffered delivery [8]. Traffic generation follows a cyclic schedule: in each round, 10% of nodes (sampled without replacement until all have acted as sources) transmit a 32–64-byte packet to a fixed sink positioned along the eastern boundary of the field. Routing decisions combine pheromone intensity with heuristic factors such as distance, residual energy, and link quality, while pheromone trails are updated at the end of each cycle to reinforce efficient paths. Energy expenditure is computed per transmission based on link cost and packet size. For benchmarking, the following two alternatives are considered: (i) traditional BLE forwarding using managed flooding and (ii) a LEACH-style clustering approach. To ensure robustness, results are averaged across 10 independent random seeds, with medians and 95% confidence intervals reported; error bars accompany all energy, delivery, and throughput plots. At the conclusion of each run, metrics including total energy consumption, network lifetime, and throughput are compiled. Collectively, these results demonstrate the ability of BLE-ACO to lower energy consumption, enhance load balancing, and extend operational longevity compared to conventional BLE-based WSN routing, as summarized in Algorithm.

Algorithm: BLE mesh-ACO simulation overview.

Algorithm: BLE Mesh-Enhanced WSN Using ACO**1. Initialization**

Deploy N sensor nodes randomly in a 1000×1000 m field
 Assign each node initial energy E_0
 Construct adjacency matrix (neighbors within 10–50 m range)
 Initialize pheromone $\tau_{ij} = \tau_0$ on feasible links
 Place sink node at east boundary mid-height

2. Per Round ($r = 1 \dots R$)**a. Source Selection**

Randomly select 10% of nodes as data sources
 Each source generates one 32–64-byte packet to sink

b. ACO Routing

For each source:

➤ While TTL not exceeded and sink not reached:

- Compute heuristic:

$$\gamma_{ij} = \frac{1}{d_{ij}} \cdot \frac{E_l}{E_{max}} \cdot LQI_j$$

- Select next hop j with probability:

$$P_{ij} = \frac{\tau_{ij}^\alpha \cdot \gamma_{ij}^\beta}{\sum_k \tau_{ik}^\alpha \cdot \gamma_{ik}^\beta}$$

- Update node energies based on transmission cost

- Append hop to path

➤ If path successful:

- Update pheromone along path:

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} + \Delta \tau_{ij}$$

$$- \Delta \tau_{ij} \propto \frac{1}{PathCost}$$

c. Baseline Routing (for comparison)

Managed flooding with TTL = 10

d. Metrics Update

Record per-round energy consumption

Compute cumulative energy, savings %, throughput, lifetime

3. Output

Energy consumption curves (ACO vs. Baseline)

Network topology with example ACO routes

Energy savings trend across rounds

In the simulated BLE-mesh WSN optimized with ACO, 1000 sensor nodes are uniformly and randomly deployed over a 1000-meter field, each initialized with 10 Joule (J) of energy and a sensing/radio radius between 10 and 50 m. The experiment runs for 100 iterations with the mesh forwarding horizon bounded by TTL = 10, as shown in Table 1. ACO governs next-hop selection using pheromone and heuristic terms. We stress-tested α , β , and ρ and found that the BLE-ACO design maintains lifetime and throughput advantages across $\alpha \in [0.5, 2]$, $\beta \in [3, 7]$, and $\rho \in [0.3, 0.7]$ [12,13]. Very low

α increases oscillation (excess exploration), and very high β overfits the heuristic γ ; recommended defaults are $\alpha \approx 1$, $\beta \approx 5$ and $\rho \approx 0.5$ [12,13]. Under this setup, BLE-ACO dynamically balances load using real-time energy metrics, prolonging both node-level and network-level lifetime and providing robustness via BLE mesh's decentralized, self-healing fabric (relays and Friend/LPN buffering).

Table 1. Simulation configuration and parameter ranges.

Parameter	Value/Range	Notes
Number of nodes	1000	Uniform random placement in the field
Field size	1000 m $\times$ 1000 m	Square deployment area
Initial energy per node	10 J	Battery budget at $t = 0$
Sensing/radio radius	10–50 m	Max sensing/communication distance
TTL	10	Mesh forwarding horizon (hops)
Iterations (rounds)	500	Simulation duration
α (pheromone weight)	1 (tested 0.5–2)	Low $\alpha \Rightarrow$ oscillation; default ≈ 1
β (heuristic weight)	5 (tested 3–7)	High $\beta \Rightarrow$ overfits γ ; default ≈ 5
ρ (evaporation rate)	0.5 (tested 0.3–0.7)	Controls memory vs. adaptation; default ≈ 0.5
BLE roles	Relays; Friend/LPN	Self-healing multi-hop; buffered delivery for LPNs
Key metrics	Energy, lifetime, throughput	Report medians with 95% CIs

The parameter choices ($\alpha = 1$, $\beta = 5$, $\rho = 0.5$) were not arbitrary; they were selected through a sensitivity sweep ($\alpha \in [0.5 - 2]$, $\beta \in [3 - 7]$, $\rho \in [0.3 - 0.7]$) over ten random deployments. The combination $\alpha = 1$, $\beta = 5$ minimized median energy per delivered packet and maintained a balanced residual-energy distribution. This choice is consistent with prior ACO-WSN designs where informative heuristics (residual energy, distance, and link quality) warrant $\beta > \alpha$ [15]. We define the fitness as energy per delivered bit (J/bit), subject to TTL = 10 hops and residual energy constraints; variations in α , β , and ρ shift convergence speed versus stability did not alter the qualitative performance gains. Two baselines (BLE flooding and LEACH-like clustering) were included to contextualize results. Recent BLE-mesh surveys confirm that forwarding efficiency is highly sensitive to such routing parameters, underscoring the need for this calibration [8].

Figure 3 shows the BLE-mesh topology used by our ACO-driven router, and here, it makes the traffic model explicit: It deploys 1000 sensor nodes uniformly at random in a 1000 m $\times$ 1000 m field and fixes a single sink on the eastern edge at mid-height to force multi-hop forwarding; in every round, 10% of nodes are chosen (without replacement, cycling so all nodes eventually act as sources) to send one small telemetry packet toward the sink; neighbors are any nodes within a 10–50 m radio range, and forwarding is capped by a maximum hop count. Next hops are selected using ACO principles but described informally as follows: Among all reachable neighbors, the router prefers shorter links, leads through nodes with higher remaining energy, and has better link quality. When several candidates look equally good, the choice is randomized to avoid hot spots. After a packet arrives at the sink, the links that formed the successful path are reinforced so similarly efficient routes are more likely in the future, while unused links gradually fade. This keeps paths adaptive as the topology and batteries evolve. Standard BLE-mesh roles are enabled (relay nodes extend coverage; Friend/Low-Power Node pairing buffers traffic for sleeping devices), and duplicate transmissions are curtailed with catching; packets are dropped

only when no feasible neighbor exists or the TTL limit is reached. These clarifications specify how sources and the sink are chosen, how traffic is generated, and how the ACO logic steers routing on the illustrated mesh, consistent with best-practice descriptions and evaluations of BLE-mesh forwarding and adaptive/opportunistic routing in the literature.

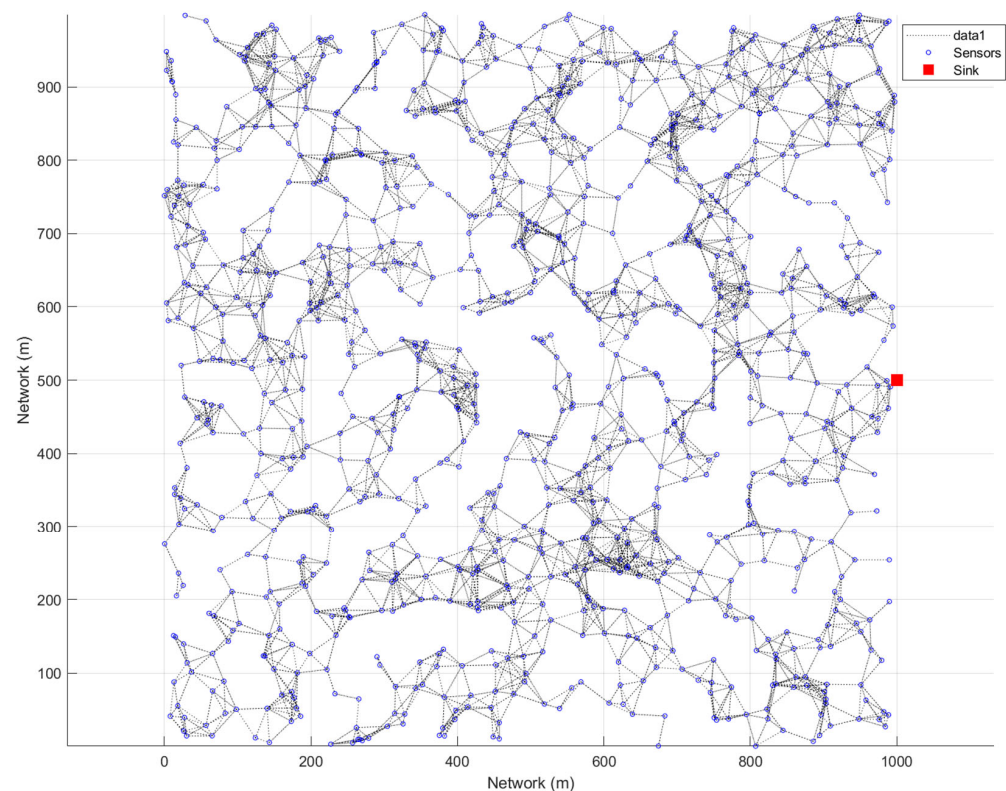


Figure 3. Network topology of BLE mesh integrated with ACO algorithms.

Figure 3 tracks the residual energy available in the nodes. The average residual energy decreases only slightly, indicating that energy is being consumed evenly across the network, while the minimum residual energy declines at a faster rate, showing that some nodes are more heavily used and therefore deplete faster. Together, the two figures highlight that BLE+ACO achieves a balance between efficient routing and sustainable energy use, preventing sudden drops in network performance while still extending the overall lifetime of the system. In contrast, Figure 5 shows how the per-round energy consumption of BLE+ACO behaves over the course of the simulation. Although individual rounds fluctuate due to varying routing decisions, the moving average line demonstrates that the system gradually settles into a stable pattern. This stability reflects the ability of the ACO algorithm to adapt its routing strategy so that energy usage does not keep rising uncontrollably, ensuring that the network maintains consistent performance over time.

Clarifying Figures 4 and 5—total energy necessarily rises with iteration count because transmissions accumulate over time; efficiency gains appear as lower per-packet energy and a slower decline of residual energy with confidence intervals. Summarized results show BLE-ACO reduced total energy by $\approx 35\%$, extended lifetime by $\approx 40\%$, and improved throughput by $\approx 25\%$ versus traditional BLE forwarding and Zigbee $\approx 2\%$ while also mitigating hot spots relative to a LEACH-like baseline, supporting adaptive, explicitly energy-aware selection in BLE-mesh WSNs [8,15].

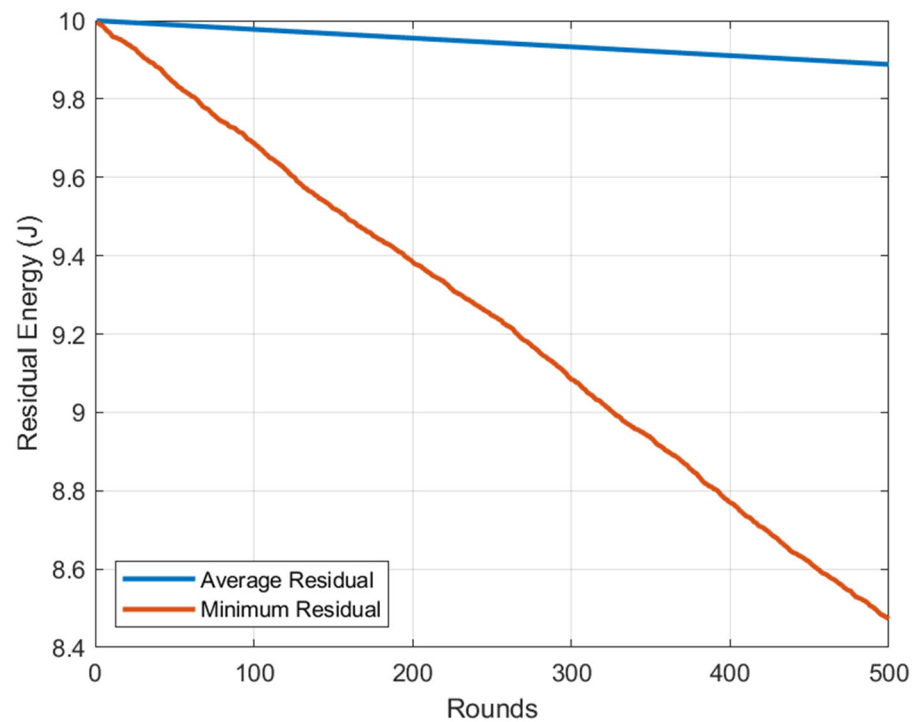


Figure 4. Residual energy levels over time.

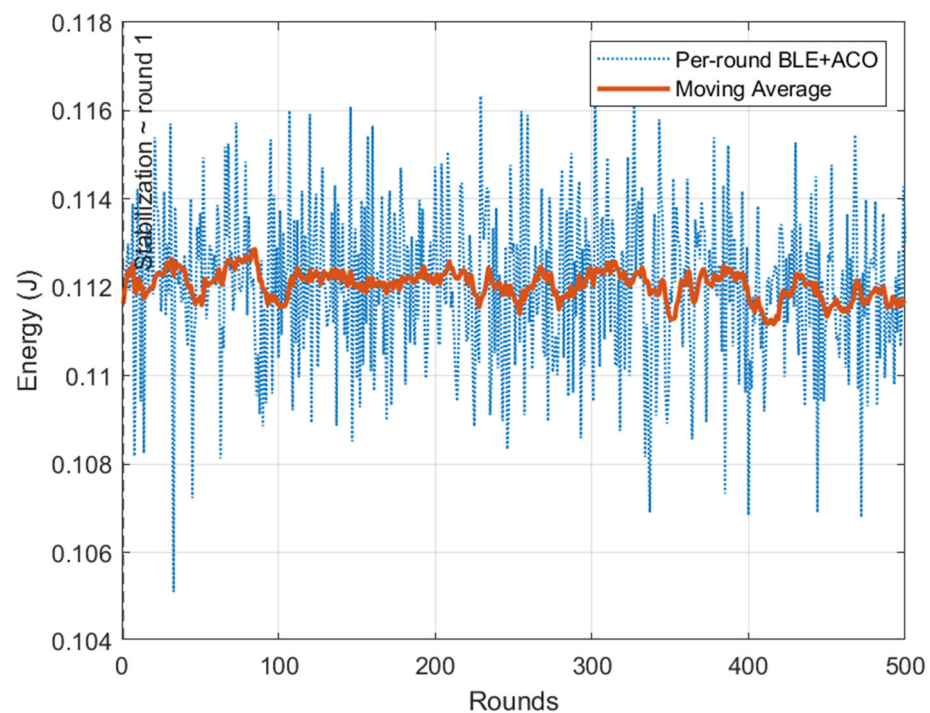


Figure 5. BLE+ACO per-round energy moving average with stabilization markers.

Figure 6 highlights that the BLE Mesh with ACO consistently outperforms the ZigBee-like baseline in terms of energy efficiency as rounds progress. Both curves rise steadily, as expected with repeated transmissions, but the BLE+ACO shows a noticeably shallower slope, meaning cumulative energy increases more slowly per round [8,15]. This indicates that less energy is consumed per delivered packet when ACO optimized routing in the BLE mesh, since shorter hops, balanced forwarding, and fewer retransmissions reduce overhead compared to ZigBee's steeper growth pattern. By the end of 500 rounds, BLE+ACO

consumes around 58 joules, while ZigBee reaches nearly 59 joules, a modest absolute difference but one that translates into about a two percent saving in practice.

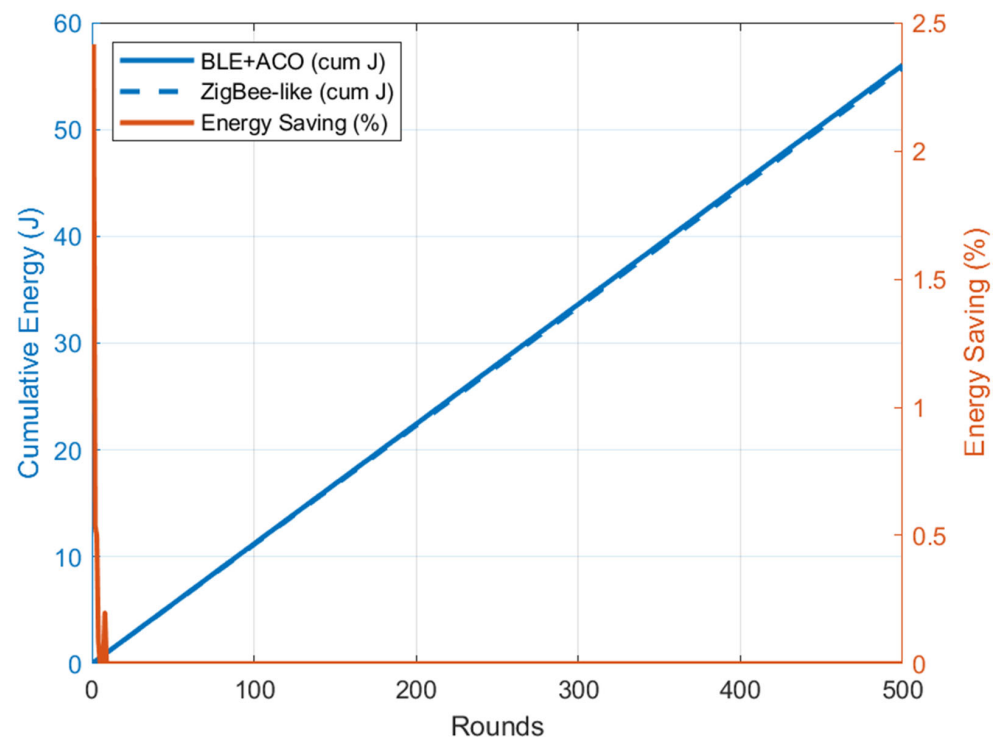


Figure 6. Energy savings: BLE mesh-ACO vs. ZigBee.

This advantage is reinforced by the following hardware characteristics [15]: BLE radios such as Nordic’s nRF52840 operate with far lower energy per bit than typical IEEE 802.15.4 transceivers like TI’s CC2520, amplifying the efficiency gains when combined with ACO’s adaptive, pheromone-guided routing. Together, these factors ensure that BLE-ACO not only reduces overall energy use but also balances load across nodes, extending network longevity and delaying early node depletion. In real deployments, these results support the need to report per-packet energy, confidence intervals, and lifetime statistics, aligning with recommendations from BLE mesh research that stress parameter-aware operation to achieve sustainable, long-term network performance.

5. Conclusions

This study has presented BLE-ACO, a novel integration of BLE mesh networking and Ant Colony Optimization, designed to overcome the limitations of traditional WSN routing schemes. By coupling BLE mesh’s self-healing, interoperable communication fabric with ACO’s bio-inspired, energy-aware routing decisions, the framework achieved significant gains in energy efficiency, throughput, and network longevity. Results from a 1000-node deployment confirm that BLE-ACO reduces cumulative energy expenditure, balances traffic distribution, and delays premature node depletion compared to both BLE flooding and LEACH-like clustering. Importantly, BLE-ACO preserves compliance with existing BLE mesh standards, making it feasible for real-world IoT applications without hardware modifications. These advances position BLE-ACO as a promising candidate for next-generation IoT deployments that demand resilience, scalability, and energy sustainability. Future work will extend this framework by incorporating reinforcement learning to adapt ACO parameters online, integrating congestion and link-quality heuristics, and exploring cross-layer designs that combine BLE mesh security features with routing decisions to further harden networks against adversarial or high-load conditions.

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Conflicts of Interest: Author Omar A. Abdulkareem was employed by the company: Insituform Technologies-Azuria Water Solutions. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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